

Internal Combustion Engine Project

Comparative study of the effects of different piston head geometries in IC engines



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INTRODUCTION

The Role of Piston Geometry:

- The bowl shape dictates the Swirl and Squish motions, which are essential for breaking up the high-pressure fuel spray.
- Enhanced turbulence reduces physical delay, leading to more complete combustion.
- Proper geometry minimizes heat loss to the cylinder walls and piston crown by centralizing the flame kernel.

Why Ansys Forte?

- It integrates specialized Chemkin-Pro solver technology for fast, accurate combustion modeling, reducing simulation time by 3-5x compared to traditional CFD

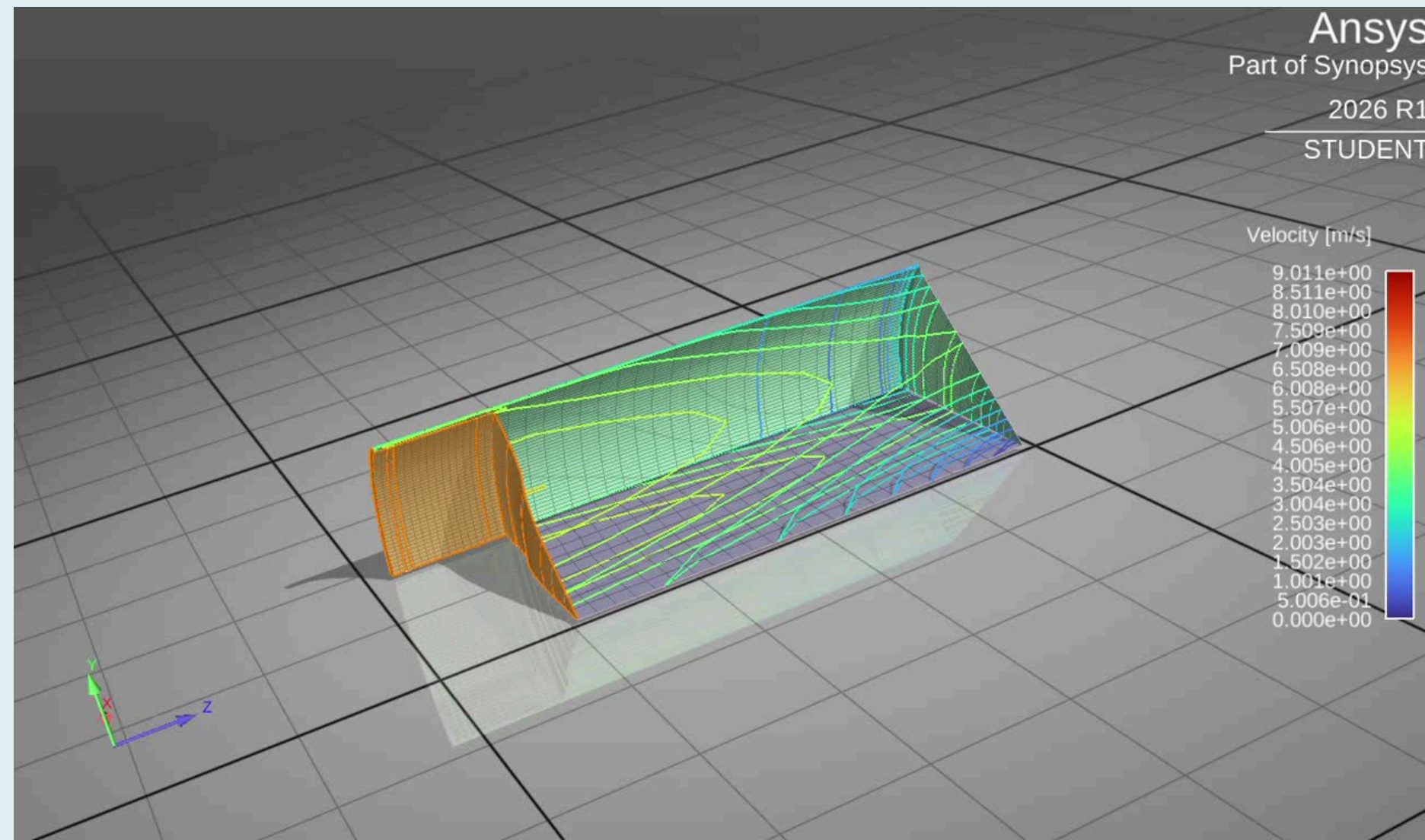


EVOLUTION OF PISTON GEOMETRIES

Geometry	Primary Benefit	Drawback
Flat / Shallow	Low heat loss	Poor mixing at high speeds
Hemispherical	Simple fabrication	High soot at low loads
Omega / Re-entrant	High Turbulence / Low Emissions	High thermal stress on bowl lip

PROBLEM STATEMENT

- In compression-ignition (CI) engines, achieving high thermal efficiency while simultaneously lowering emissions is a major technical hurdle.
- Conventional or flat piston designs often suffer from poor Air Utilization.
- There is a critical need to evaluate how subtle changes in the "bowl lip" or "bottom curvature" of a piston can optimize the combustion rate without increasing thermal stress on the engine components.



OUR SOLUTION

- We have used Ansys Forte for geometry formation and simulation purposes.
- We generated the sector mesh using the sector mesh generator in Forte.
- All the engine parameters were given during sector mesh generation.
- We have done simulations on 2 piston head geometries (Flat and circular dish type).
- All the required settings were made in Ansys Forte for a successful simulation.
- We obtained results for the simulated engines and analyzed the results.

NOTE: We have used CI engines in the simulation.

The results may not be perfect, but the effort was maxed out.

PISTON GEOMETRIES UNDER STUDY



- **Design Characteristics:** A simple, flat top surface with no recessed bowl.
- **Fluid Dynamics:** Relies almost entirely on the intake port design for swirl. It lacks Squish motion (the radial inward flow of air as the piston approaches TDC).
- **Performance Expectation:** Often results in higher "wall wetting" (fuel hitting the liner) and slower air-fuel mixing, leading to higher soot at high loads.



- **Design Characteristics:** Features a shallow, bowl-shaped indentation in the piston crown.
- **Fluid Dynamics:** Significantly enhances Squish and Tumble ratios. The "dish" acts as a combustion chamber that contains the fuel spray, concentrating the energy at the center.
- **Performance Expectation:** Improved turbulence leads to faster combustion, higher peak pressures, and a more homogeneous temperature distribution.

SIMULATION SETUP

Numerical Framework:

- Solver: ANSYS Forte (Sector Mesh)
- Chemistry Solver: CHEMKIN Chemistry Set (for precise ignition and emission kinetics).
- Turbulence Model: RANS RNG k-epsilon (Optimized for high-strain flows and swirling motion in engines).

Engine Geometry & Initial Conditions:

- Boundary Conditions: Isothermal wall temperatures assigned to the Piston, Head, and Liner.
- Initial Conditions: Swirl Ratio of 0.3 with a Swirl Profile Factor of 3.11.
- Kinematics: Defined by specific Stroke and Connecting Rod lengths to dictate piston motion.

SIMULATION SETUP

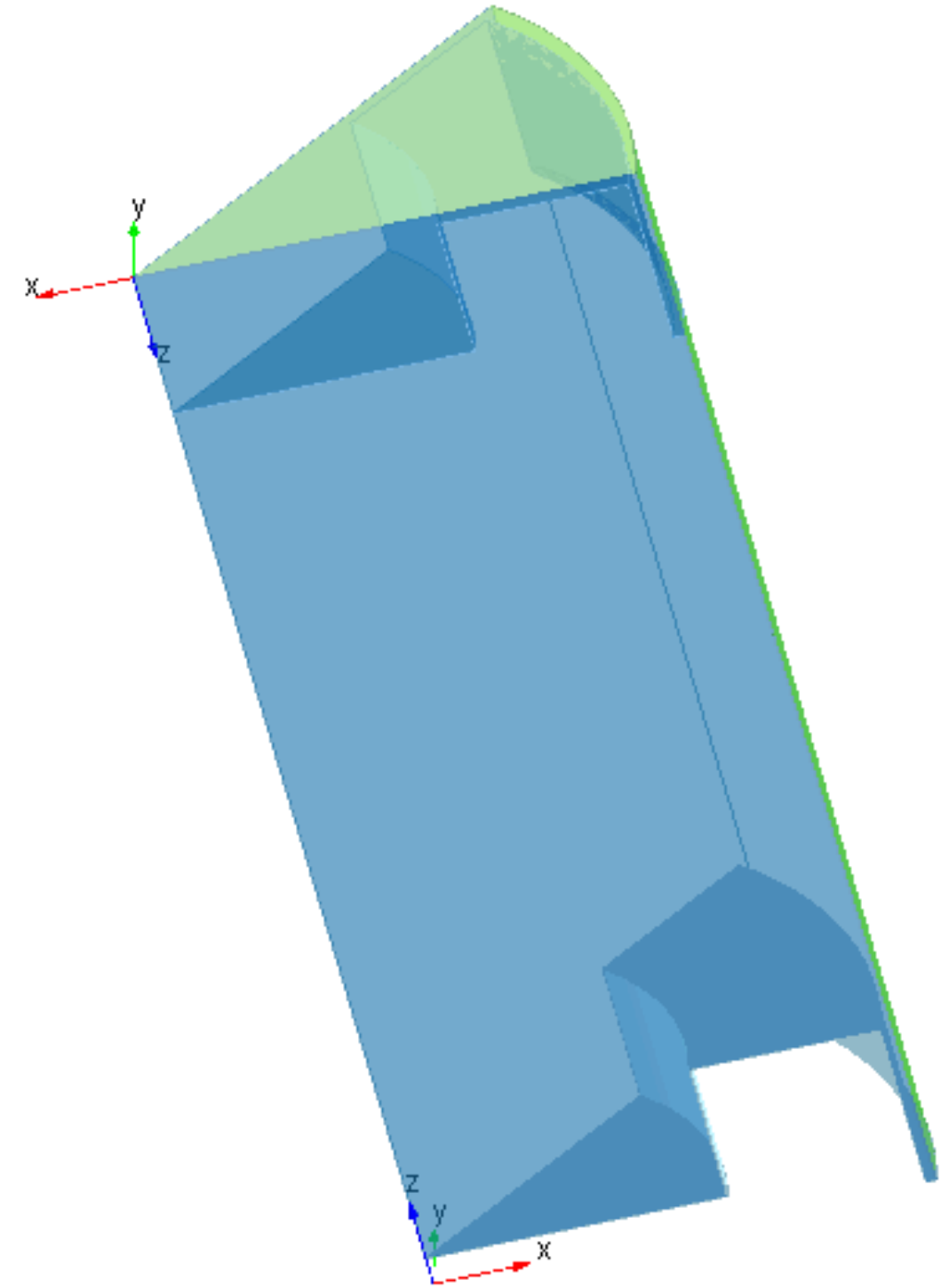
Fuel Injection Parameters (Solid Injector Model):

- Injector Location: Centrally located at the sector edge (represents a multi-hole nozzle).
- Nozzle Diameter: 150 micron.
- Spray Dynamics: 15° Cone Angle with 4000 parcels for statistically significant spray distribution.
- Injection Timing: Starts at -15° BTDC (Before Top Dead Center).
- Injection Duration: 10° Crank Angle (CA).
- Mass & Velocity: 30 mg total mass injected using a Sine Velocity Profile for realistic mass-flow representation.

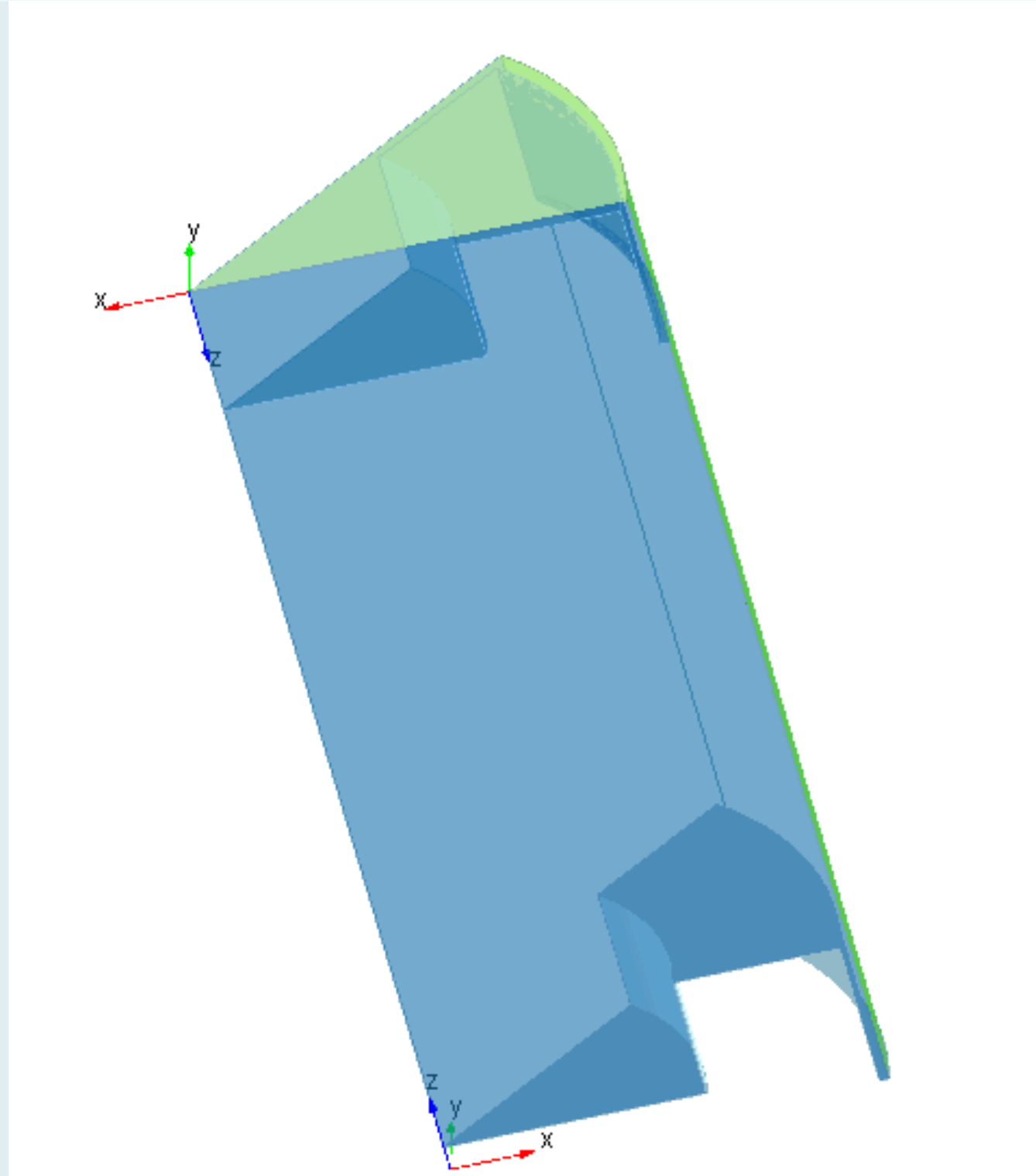
SIMULATION SETUP

Engine Parameters and mesh details:

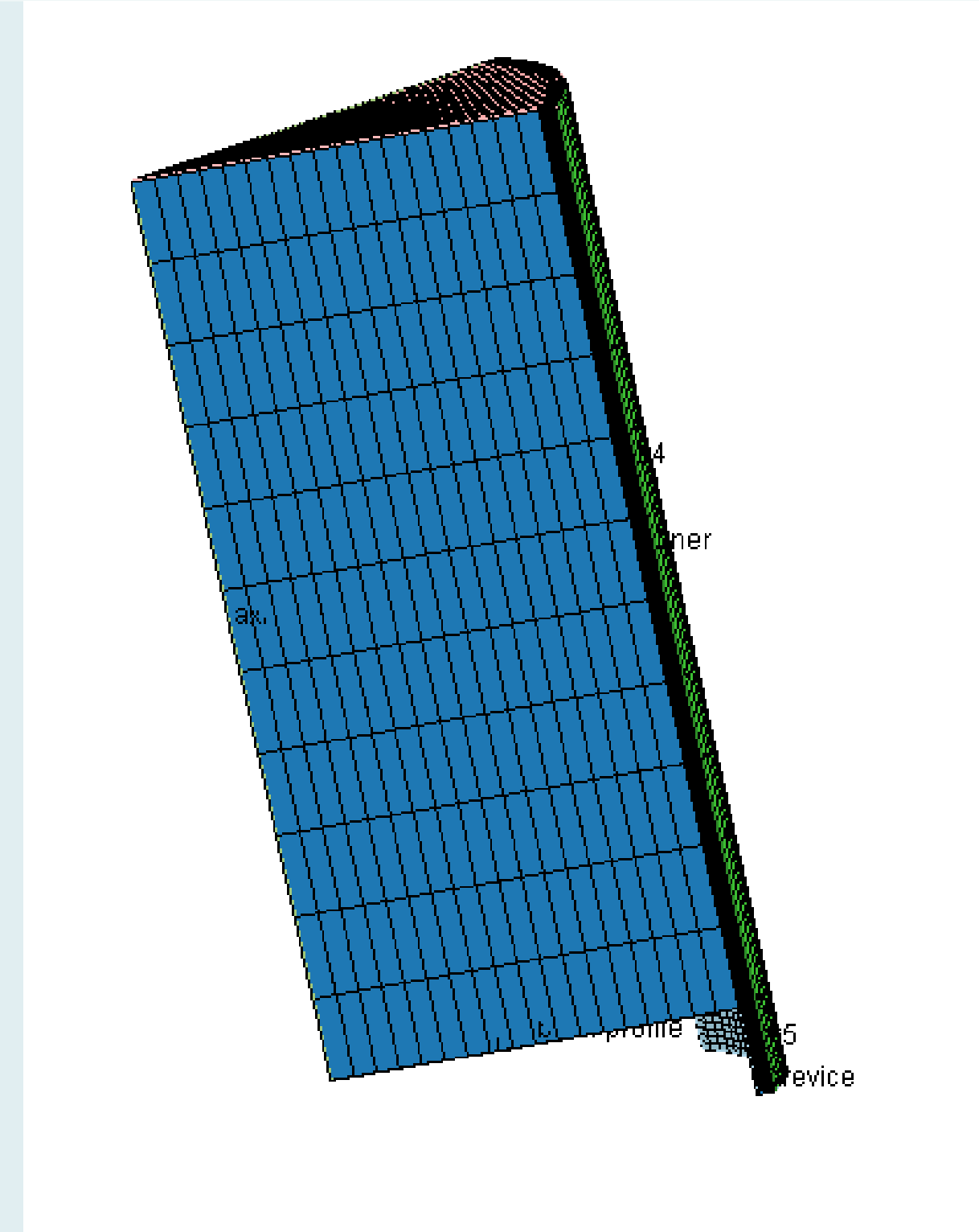
- Bore diameter: 10 cm
- Stroke length: 10 cm
- Connecting rod length: 175 cm
- We used topology 3 available in the Forte mess generator for generation of mesh.



SIMULATION SETUP

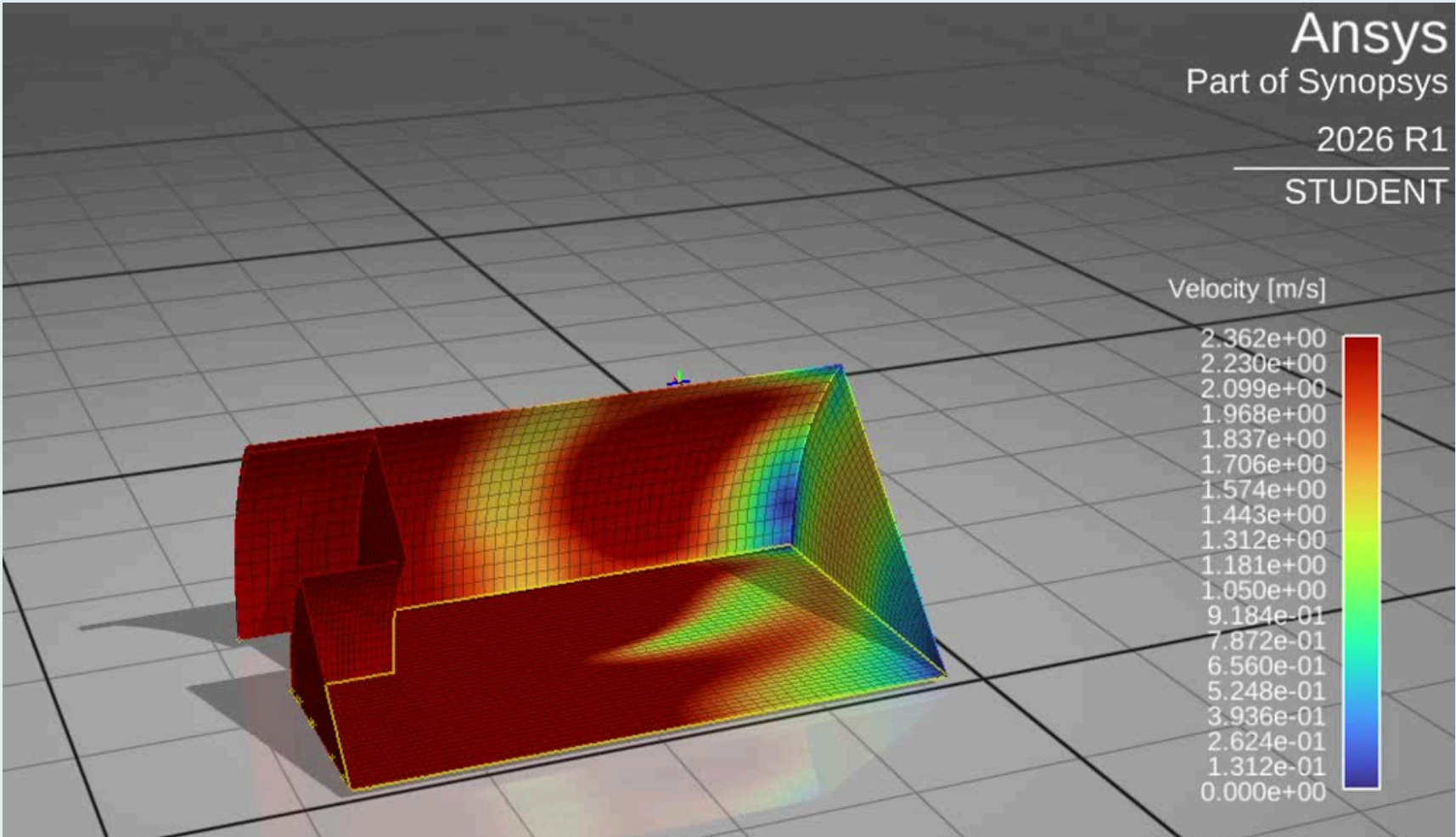


Dish Type Piston head

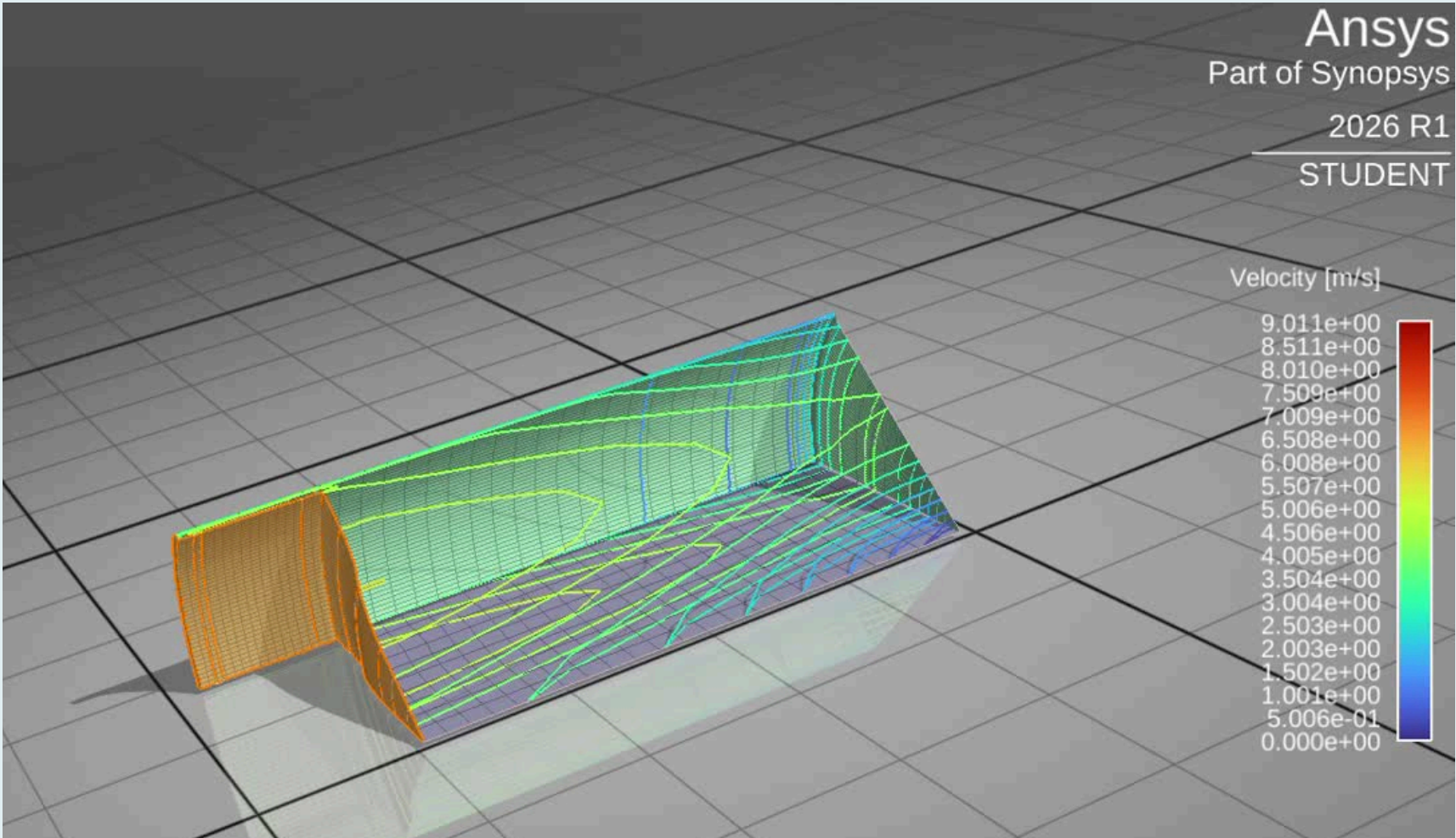


Flat Type Piston head

SIMULATION RESULTS

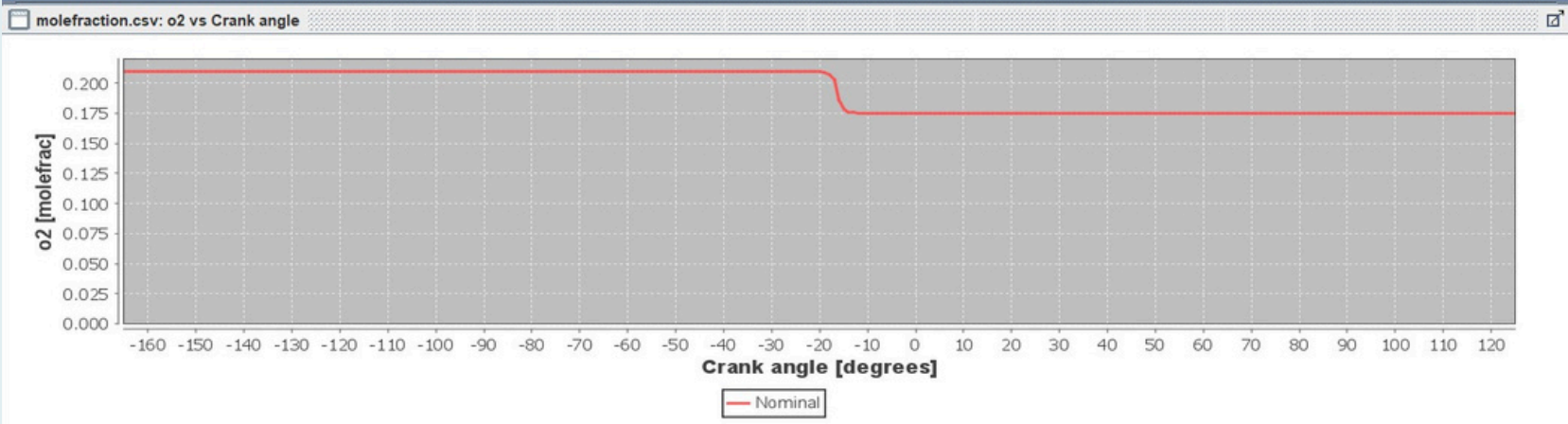
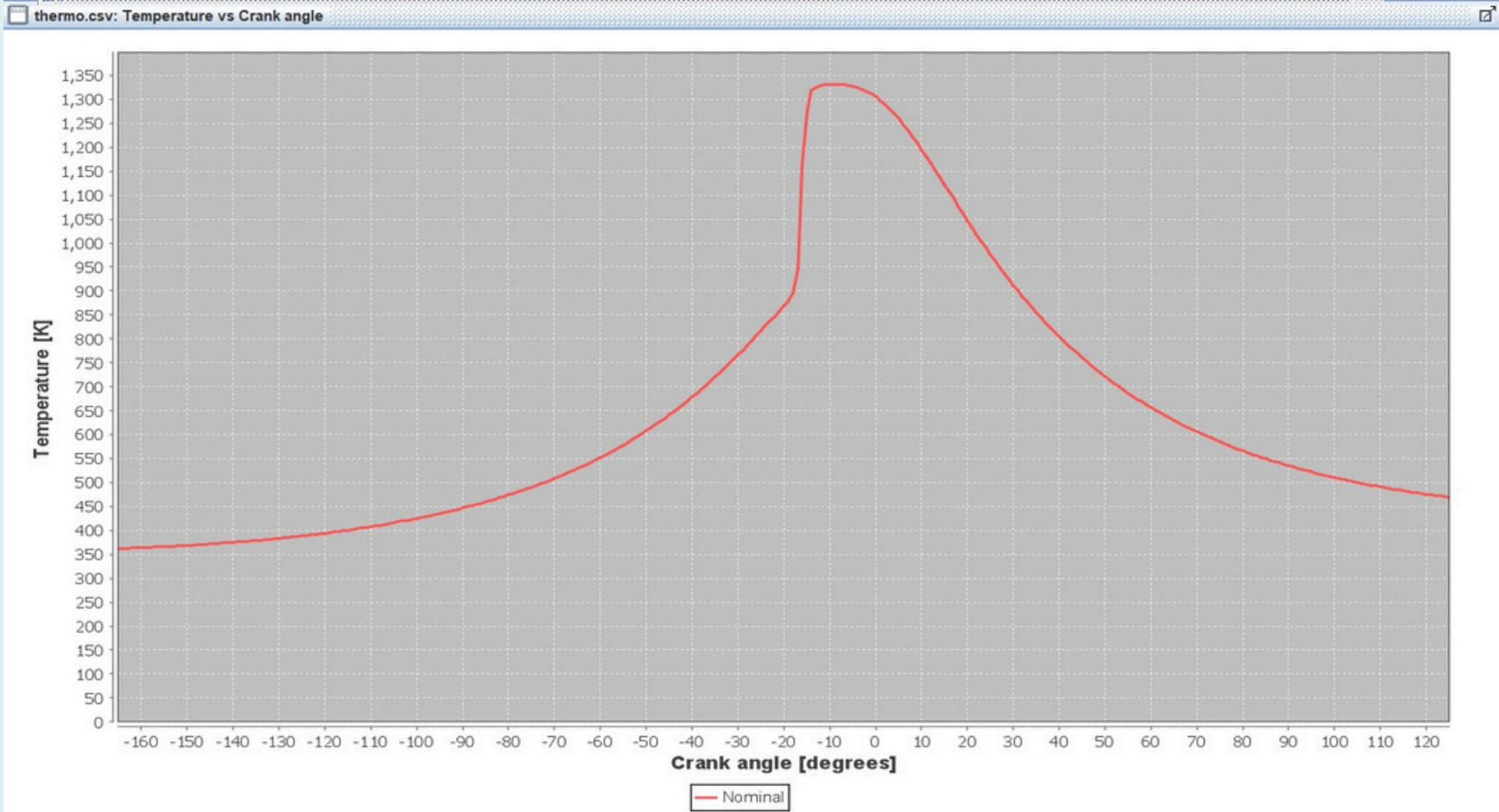
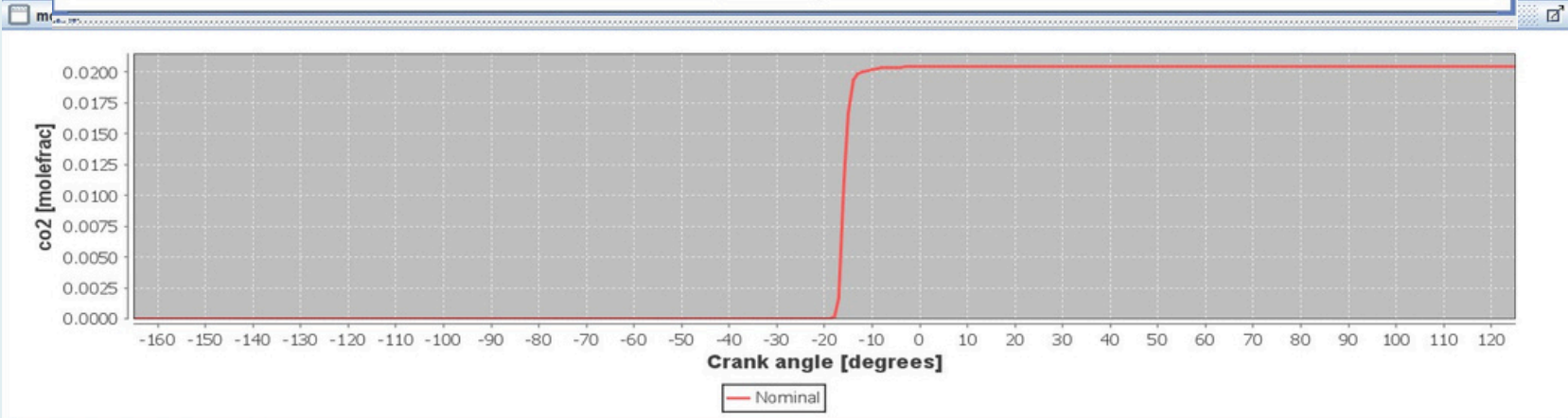
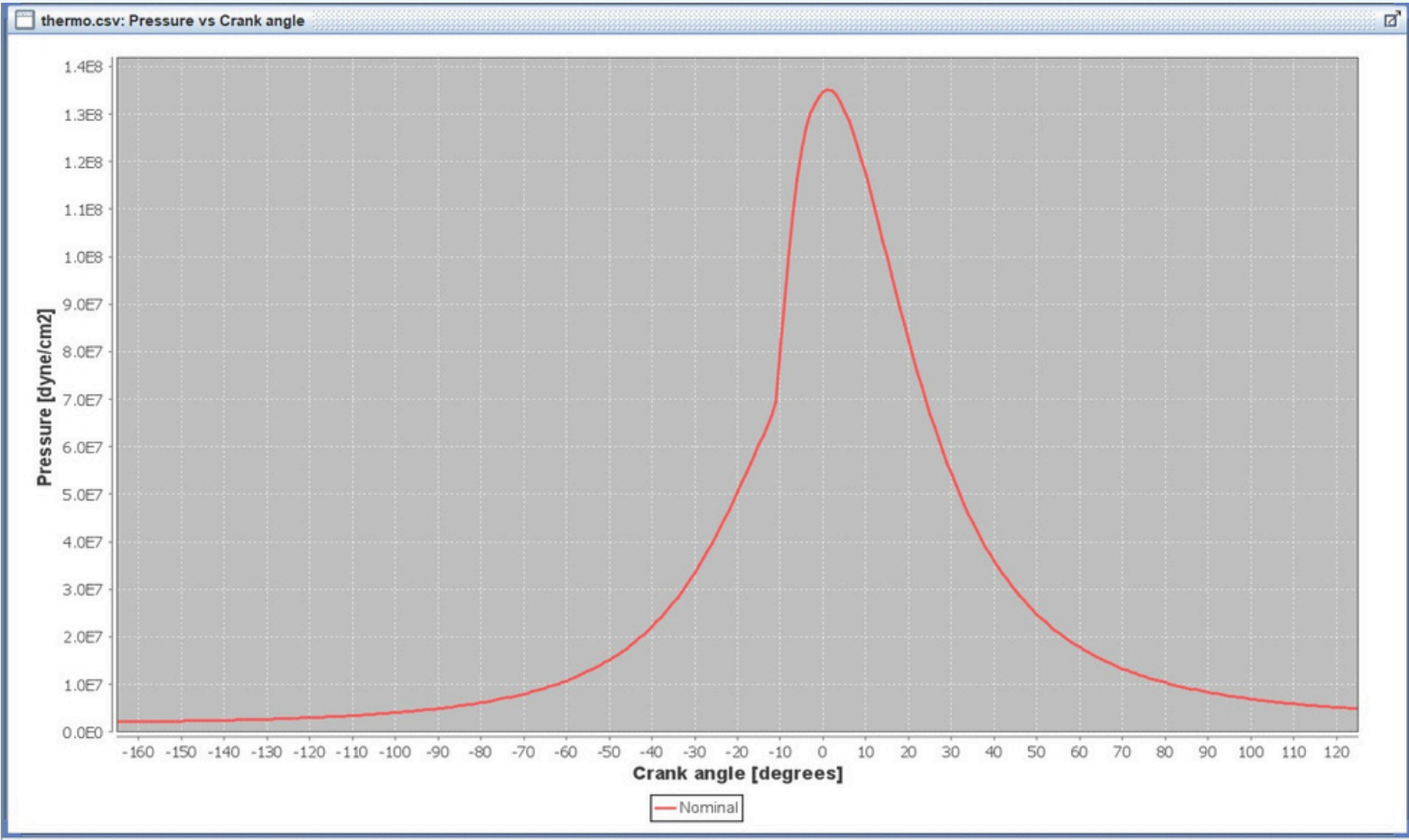
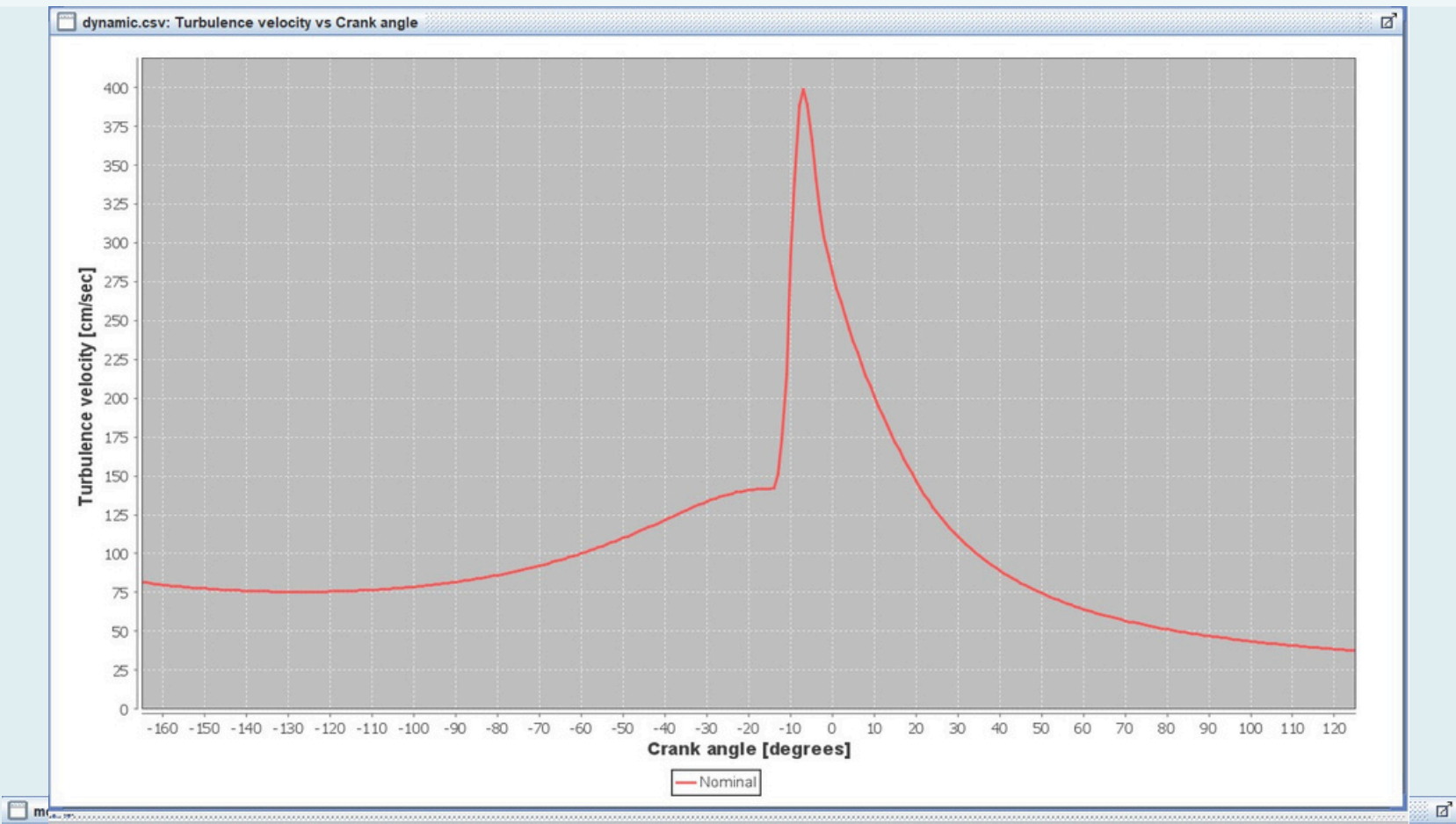


Dish Type Piston head

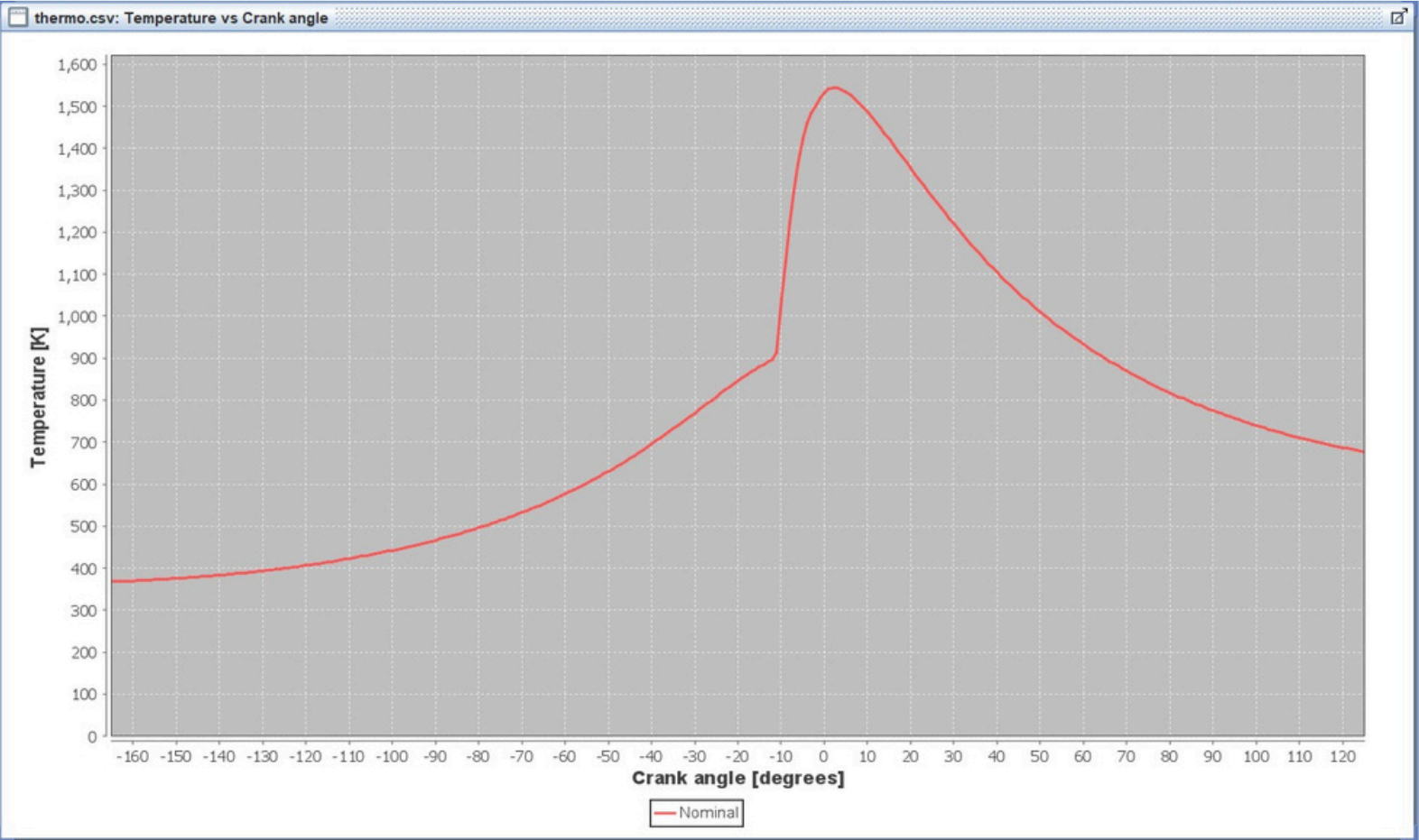
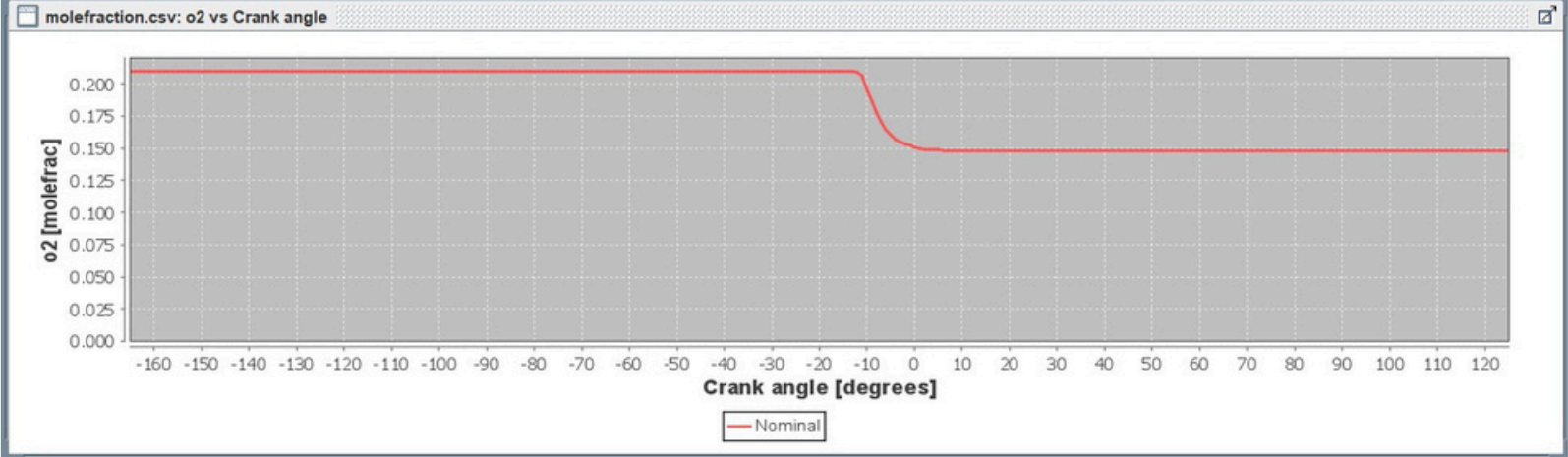
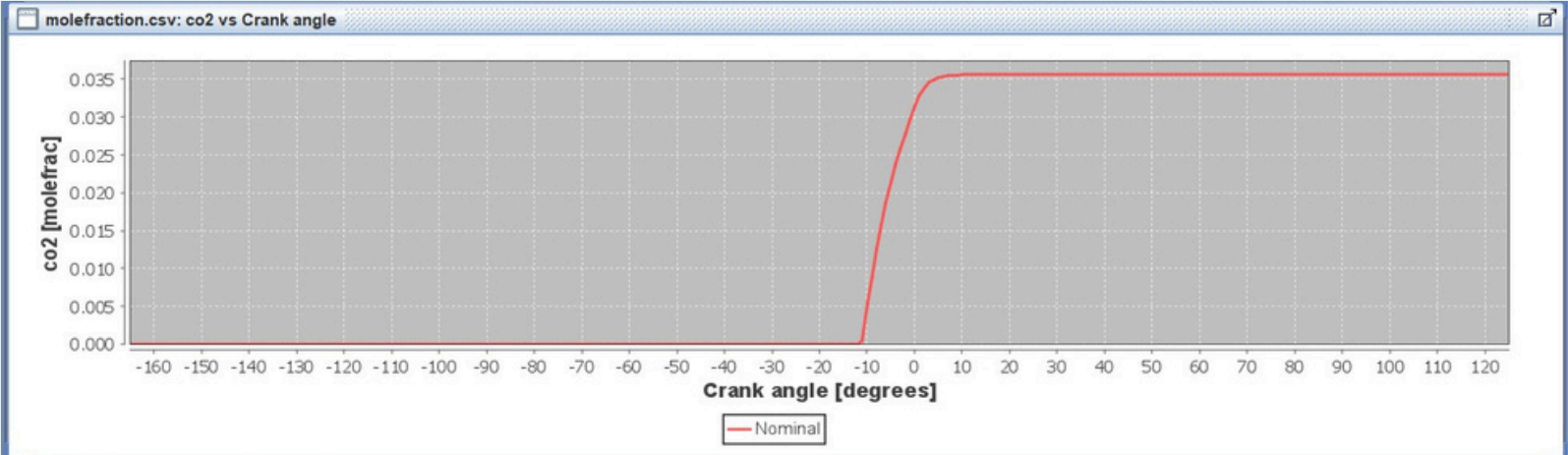
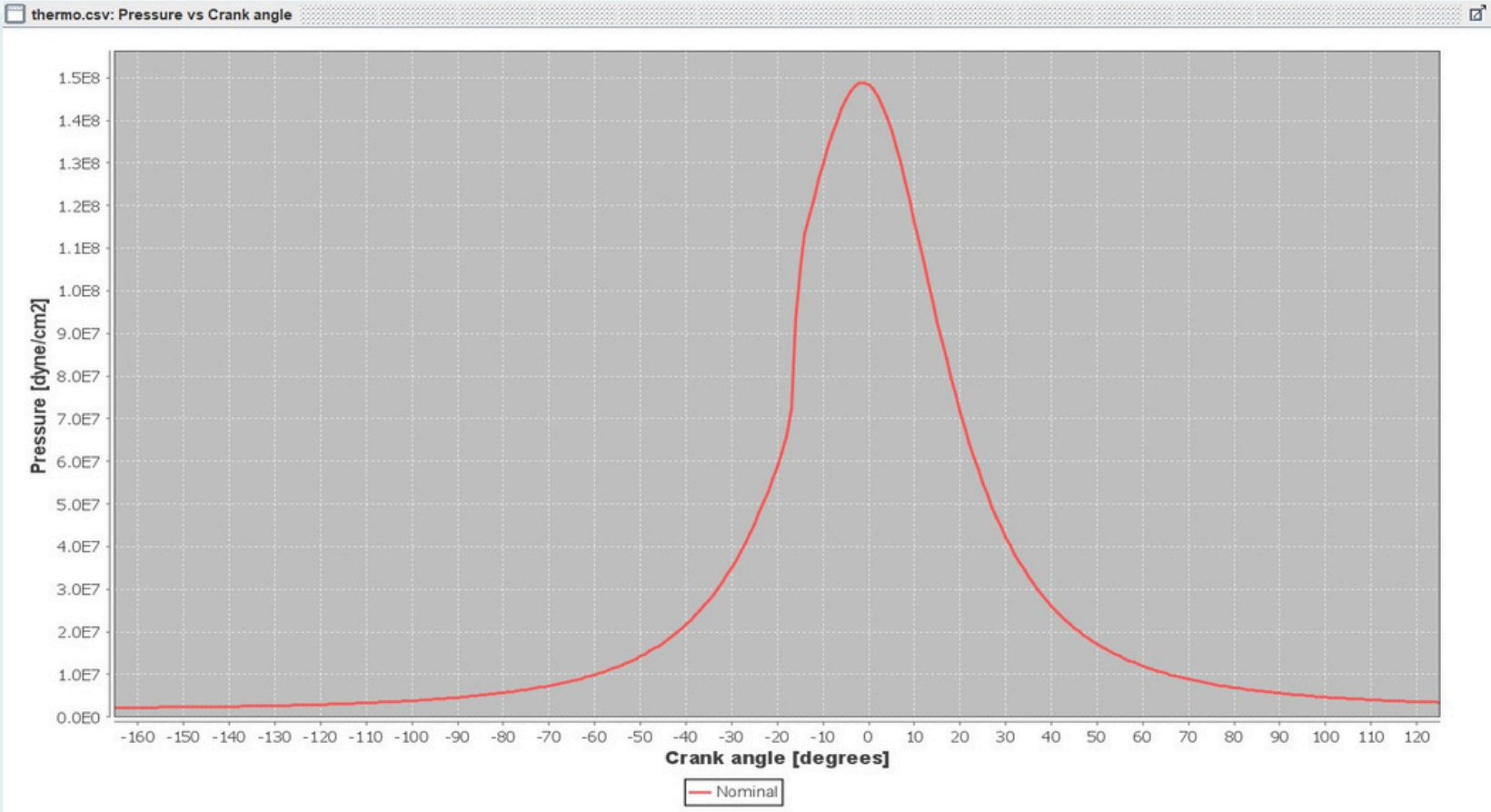
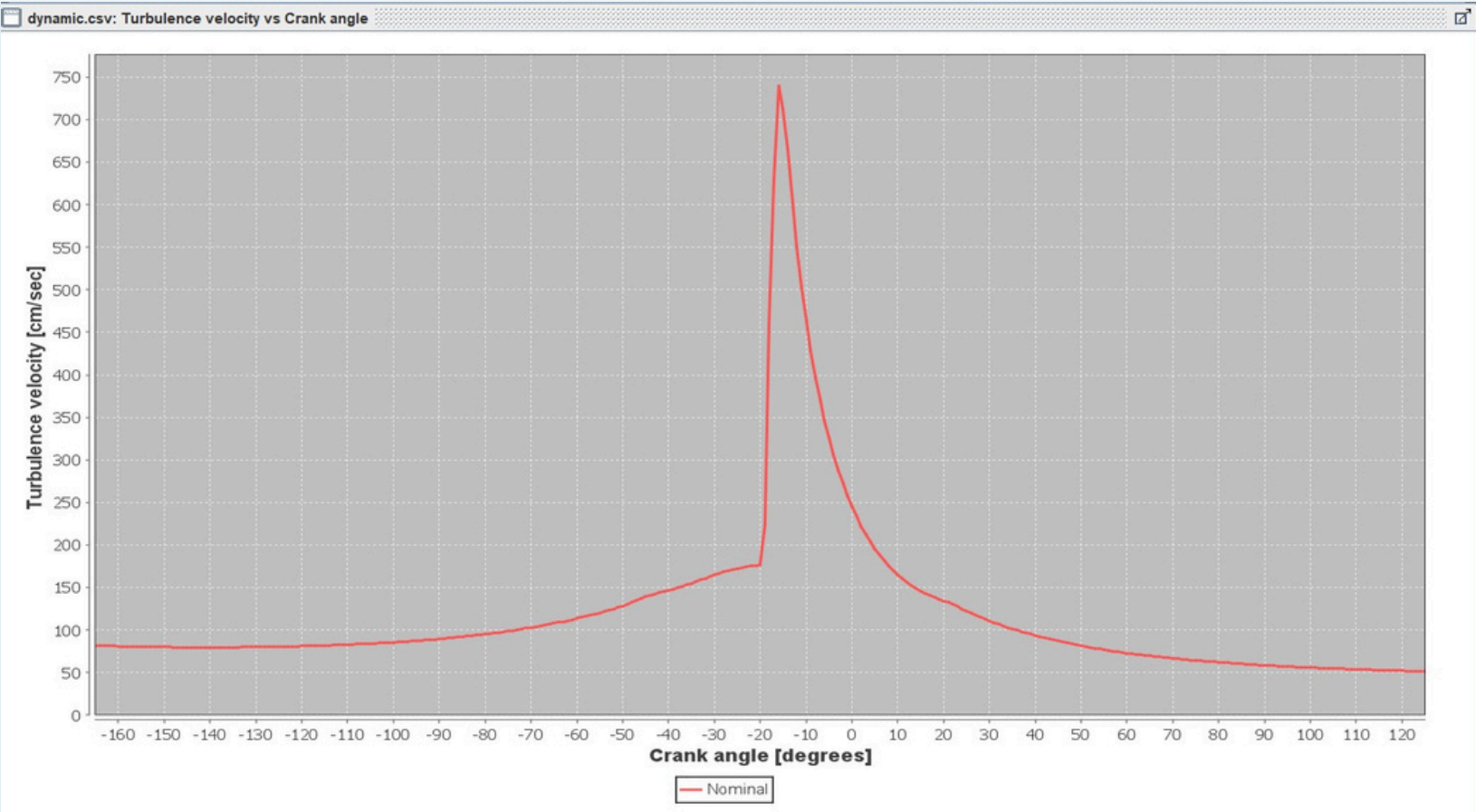


Flat Type Piston head

SIMULATION RESULTS (FLAT TYPE)



SIMULATION RESULTS (DISH TYPE)



CHALLENGES

1. Simulation Convergence Errors

Attempts to simulate 6–7 different geometries resulted in recurring solver errors and mesh instabilities at complex bowl boundaries. We pivoted to focus exclusively on Flat and Dish-type heads to ensure data reliability and project completion.

2. Computational Limitations

The high CPU and RAM demands of CHEMKIN chemistry solvers exceeded the capabilities of our available hardware. Without access to high-performance computing (HPC), we were limited in the number of iterations and mesh density we could achieve.

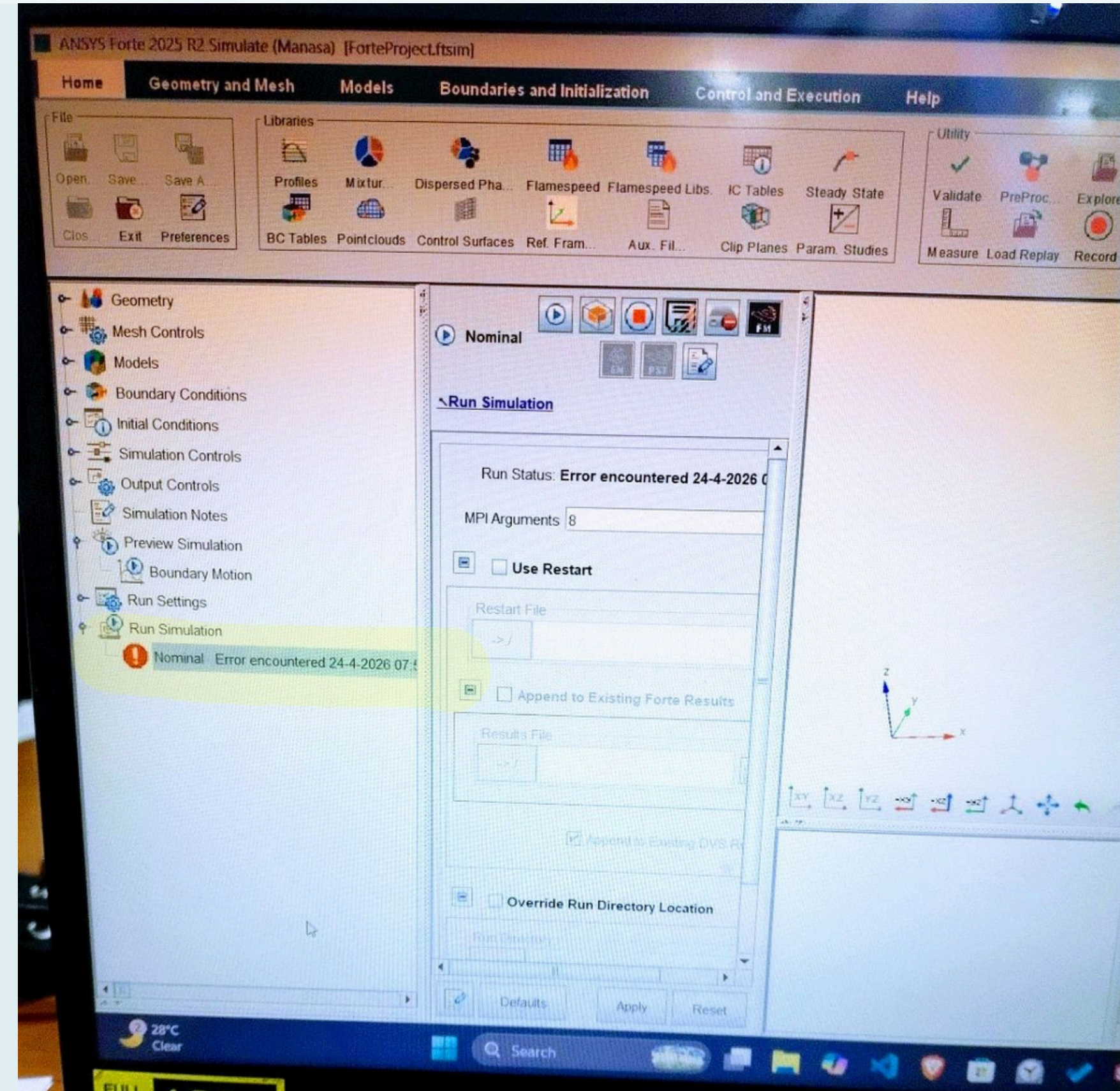
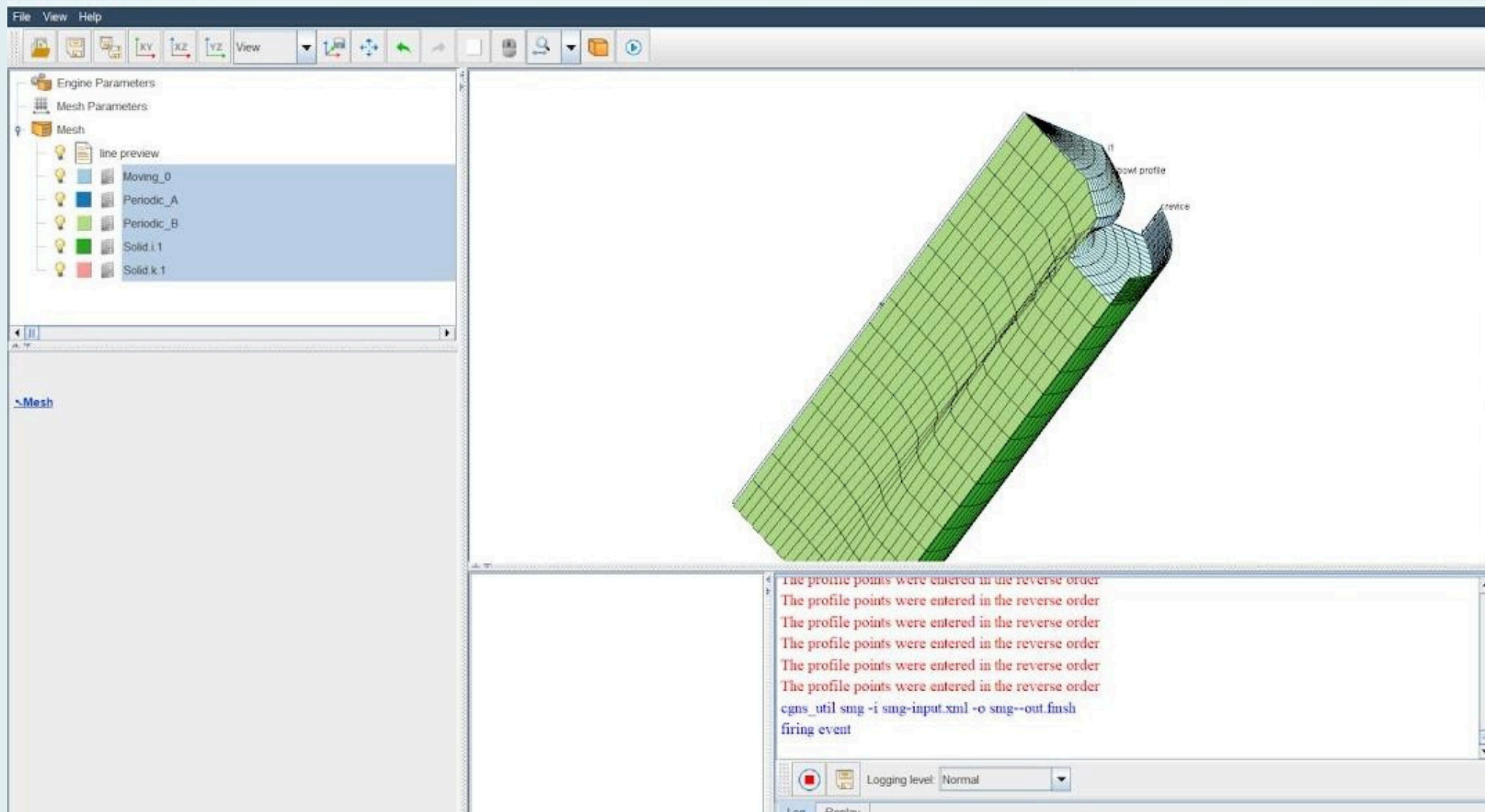
3. Resource Accessibility

A lack of detailed, open-source technical data for non-standard piston shapes made troubleshooting and validation difficult. Much of the industry-standard benchmark data remains proprietary, creating a significant information gap for student research.

4. Technical Learning Curve

Navigating the complex workflow of ANSYS Forte presented a steep learning curve for our inexperienced team. The majority of our project timeline was consumed by the "trial and error" phase of mastering the software rather than data optimization.

CHALLENGES



CHALLENGES

ANSYS Forte 2026 R1 Simulate (rudra) [ForteProject.fsim]

Home Geometry and Mesh M

Unserved Project - Workbench

File View Tools Units optiSLang Extensions Jobs Help

Project

Import... Reconnect Refresh Project Update Project ACT Start Page

Toolbox

Analysis Systems

- Coupled Field Harmonic
- Coupled Field Modal
- Coupled Field Static
- Coupled Field Transient
- Eigenvalue Buckling
- Electric
- Explicit Dynamics
- Fluid Flow (CFX)
- Fluid Flow (Fluent with Fluent Meshing)
- Fluid Flow (Fluent)
- Fluid Flow (Polyflow Classic)
- Fluid Flow (Polyflow)
- Harmonic Acoustics
- Harmonic Response
- Hydrodynamic Diffraction
- Hydrodynamic Response
- LS-DYNA
- LS-DYNA Acoustics
- LS-DYNA Restart
- Magnetostatic
- Modal
- Modal Acoustics
- Motion
- Particle Dynamics (Rocky)
- Random Vibration
- Response Spectrum
- Rigid Dynamics
- Speos
- Static Acoustics
- Static Structural

Project Schematic

1 Forte

2 Setup

3 Solution

Forte

Ansyes Workbench

Update failed

OK

Messages

	A	B	C	D
1	Type	Text	Association	Date/Time
2	Error!	Update failed		24-04-2026 20:25:42
3	Error!	Update failed		24-04-2026 20:24:54
4	Error!	Update failed		24-04-2026 20:23:49
5	Error!	Update failed		24-04-2026 20:23:30
6	Error!	Update failed due to Physics validation errors: Initial Conditions->Region 1 Initialization (Main)->Velocity->Velocity: Piston Boundary is missing value		24-04-2026 20:15:40
7	Error!	Update failed due to Physics validation errors: Initial Conditions->Region 1 Initialization (Main)->Velocity->Velocity: Piston Boundary is missing value		24-04-2026 20:14:56

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Nominal Running for 0 min

Busy

No HPC Platform Services Connection

Job Monitor...

Show Progress

Hide 6 Messages

Log

Replay

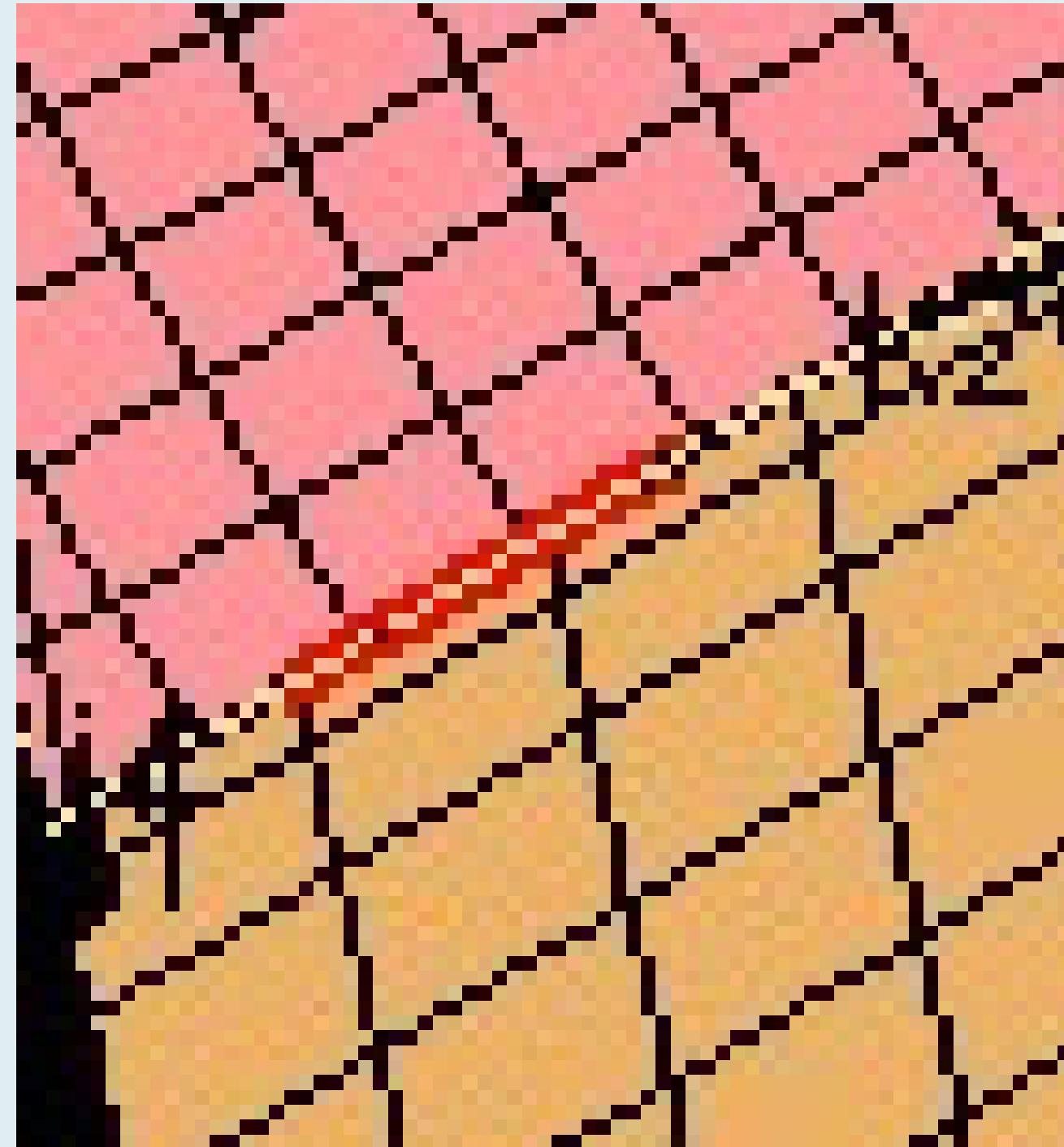
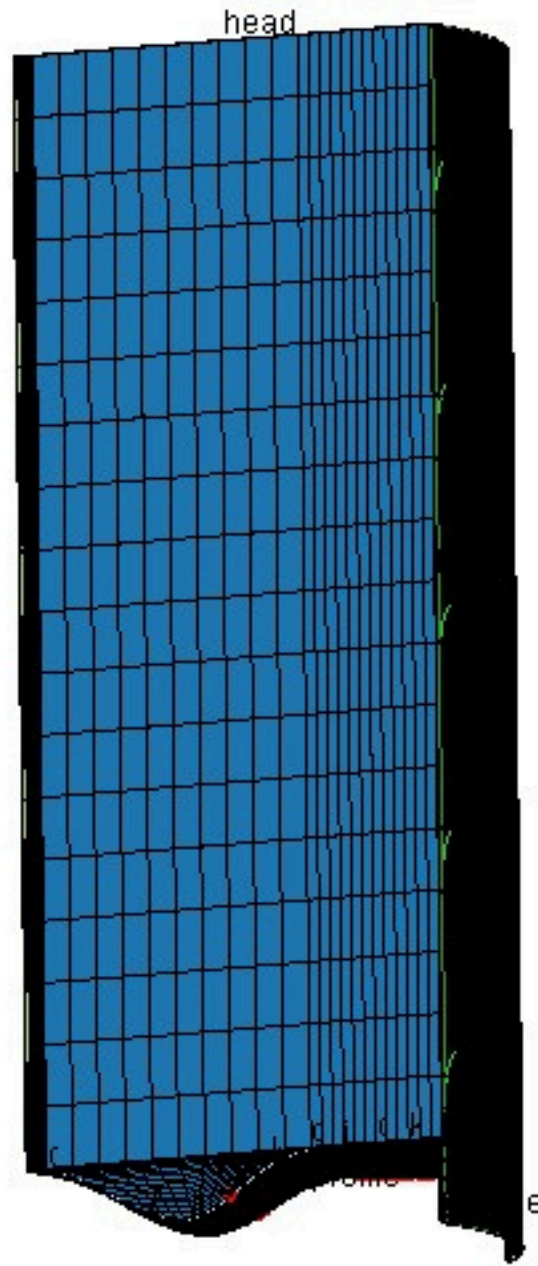
Defaults

Apply

Reset

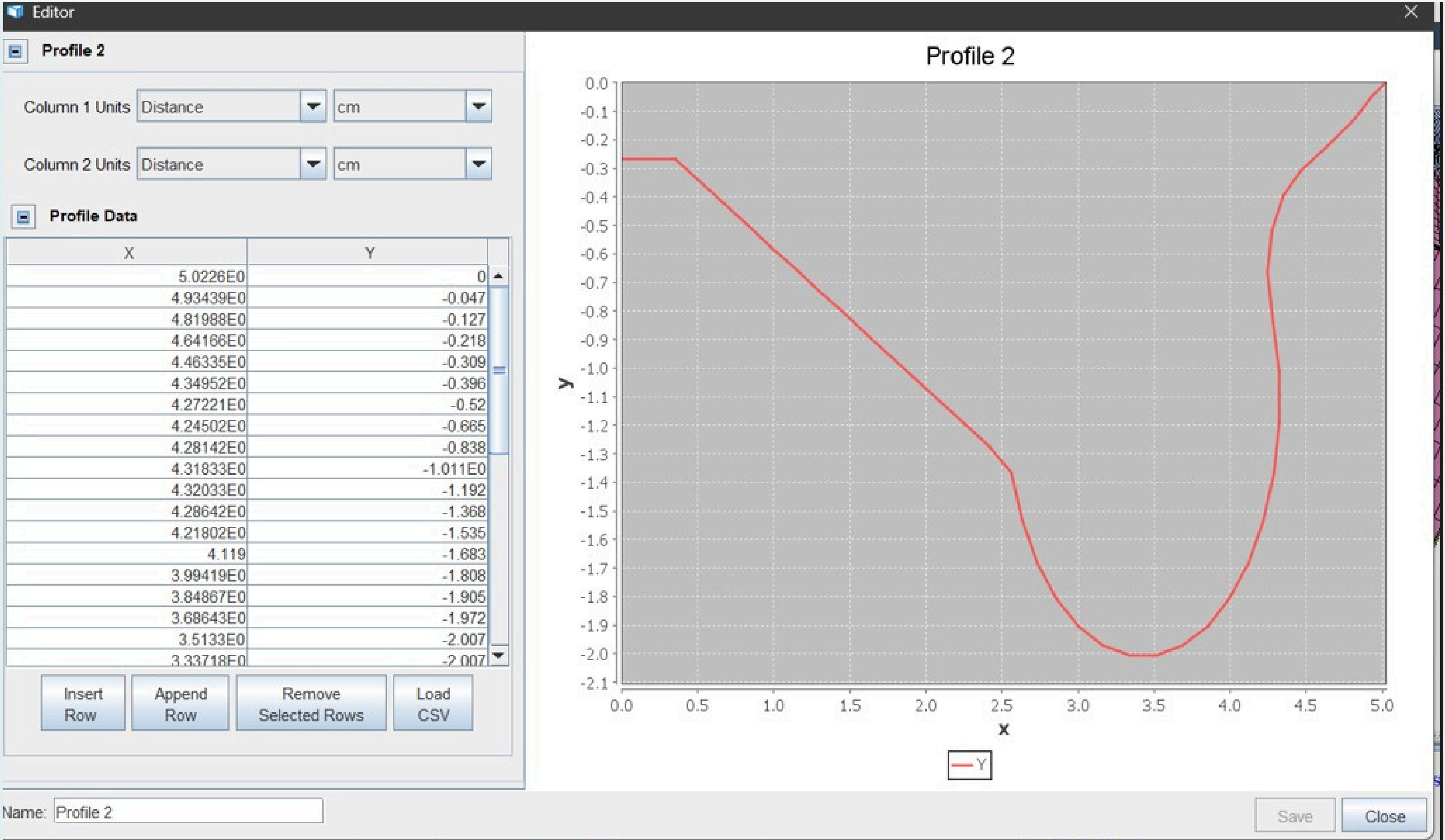
Ansyes 2026 R1 STUDENT

CHALLENGES



invalid faces error

CHALLENGES



CONCLUSION

The analysis of the Dish-type piston geometry confirms superior combustion efficiency and thermodynamic performance through the following findings:

- **Superior Turbulence:** The peak Turbulence Velocity of 400 cm/sec confirms that the dish geometry effectively generates the high-intensity squish and swirl required for rapid air-fuel mixing.
- **Robust Thermodynamic Profile:** The simulation achieved a peak pressure of 135 bar and a peak temperature of 1550 K, indicating high thermal efficiency and a well-timed heat release.
- **Rapid Chemical Conversion:** The sharp transition in O₂ and CO₂ mole fractions at -10° CA validates that the geometry promotes near-complete combustion with minimal ignition delay.
- **Final Result:** The dish-type head successfully concentrates combustion at the cylinder center, optimizing power output while managing the turbulence-driven mixing process.

THANK YOU